

Strategy 1 (Trend following): This strategy uses the ADX (average directional index), a measure of stock momentum, and a 20-day exponential moving average, which measures recent average performance. A buy order is triggered if $ADX > 25$ and the current price is greater than the EMA 20. The purpose of this strategy is to identify the beginning of upward trends, as an increase in price ($price > EMA\ 20$) combined with a large ADX value (which indicates the building of momentum) can be a signal that the stock price will go up in the future. The strategy triggers a sell order when the current price drops below the EMA 20, at which point the anticipation is that the stock will no longer rise in value.

Strategy 2 (Mean reversion): This strategy uses the RSI, which measures the extent to which a stock is overbought or oversold, as well as a SMA 14 which is the average closing price over the past 14 days and Bollinger bands, which are standard deviations of stock price. A buy order is triggered if $RSI < 50$ and the closing price crosses the midpoint between the lower Bollinger band and the SMA 14 moving upwards. The timing of this entry into a position intends to take advantage of a stock reverting to the mean (while limiting the influence of false signals) by identifying underbought, underperforming stocks that are starting to trend back to their normal levels. The strategy triggers a sell order when $RSI > 70$, the stock price reaches the upper Bollinger band, or (in order to cut losses if the stock does not continue to rise) when the stock value crosses the midpoint between the lower Bollinger band and the SMA 14 moving downwards.

Strategy 3 (Volatility Breakout): This strategy uses the ATR (average true range), which is a measure of the volatility of a stock based on the range of its recent highs and lows, the ATR-SMA 20 which is the simple moving average of the ATR over the past 20 days, the Williams %R, a recent momentum measurement, and the EMA 20. A buy order is triggered if $ATR < ATR-SMA\ 20$ and $Williams\ \%R > -50$. This combination of signals is designed to recognize when a stock is behaving in a less volatile fashion than usual (which often precedes a high volatility period), while the price is moving upwards. A sell order is triggered if the Williams %R drops below -50 or if the closing price drops below the EMA 20, which pulls the strategy out of trades once the volatility/value stops increasing.

(Performance tables for 4 of the 8 stocks tested - all 8 are visible on the streamlit app)

JPM

Performance metrics

Analytics Metric	Buy & Hold (Baseline)	Strategy 1 (Trend Following - EMA20)	Strategy 2 (Enhanced Mean Reversion)	Strategy 3 (Volatility Breakout)
Total Return	117.76%	-4.92%	38.52%	22.41%
CAGR	16.94%	-1.01%	6.77%	4.15%
Volatility	24.56%	7.55%	12.86%	13.20%
Sharpe Ratio	0.76	-0.10	0.57	0.37
Sortino Ratio	1.10	-0.12	0.87	0.52
Max Drawdown	-40.65%	-20.27%	-17.37%	-20.10%
Win Rate	0.00%	25.00%	54.55%	39.39%

SPY

Performance metrics

Analytics Metric	Buy & Hold (Baseline)	Strategy 1 (Trend Following - EMA20)	Strategy 2 (Enhanced Mean Reversion)	Strategy 3 (Volatility Breakout)
Total Return	72.26%	16.12%	0.04%	27.54%
CAGR	11.56%	3.05%	0.01%	5.01%
Volatility	17.21%	5.34%	10.59%	9.02%
Sharpe Ratio	0.72	0.59	0.05	0.59
Sortino Ratio	1.04	0.82	0.07	0.82
Max Drawdown	-25.36%	-12.48%	-19.88%	-10.12%
Win Rate	0.00%	41.18%	47.06%	41.07%

BA

Performance metrics

Analytics Metric	Buy & Hold (Baseline)	Strategy 1 (Trend Following - EMA20)	Strategy 2 (Enhanced Mean Reversion)	Strategy 3 (Volatility Breakout)
Total Return	-7.44%	40.01%	-10.20%	22.42%
CAGR	-1.54%	7.00%	-2.14%	4.15%
Volatility	36.59%	11.37%	21.87%	17.99%
Sharpe Ratio	0.14	0.65	0.01	0.32
Sortino Ratio	0.20	0.96	0.02	0.45
Max Drawdown	-51.64%	-13.77%	-52.71%	-23.49%
Win Rate	0.00%	40.00%	50.00%	30.88%

XOM

Performance metrics

Analytics Metric	Buy & Hold (Baseline)	Strategy 1 (Trend Following - EMA20)	Strategy 2 (Enhanced Mean Reversion)	Strategy 3 (Volatility Breakout)
Total Return	122.06%	20.00%	27.44%	-14.67%
CAGR	17.40%	3.73%	5.00%	-3.14%
Volatility	26.75%	8.63%	12.95%	14.82%
Sharpe Ratio	0.73	0.47	0.44	-0.14
Sortino Ratio	1.05	0.74	0.66	-0.19
Max Drawdown	-20.65%	-9.45%	-13.14%	-37.38%
Win Rate	0.00%	55.56%	51.52%	29.58%

Commentary:

In general, these three strategies have different approaches that allow them to capitalize - or be crushed - during changing market conditions.

The trend reversal strategy (1) performs well in trending markets which experience extended periods of growth, which the ADX+EMA takes advantage of, or periods of decline, which the strategy avoids investing into. This is seen in the performance metrics for BA. Despite the overall value of the stock decreasing over the 5 year time period (-7.44%), strategy 1 saw a 40.01% portfolio growth, due to the long stretches of predictable movement. This strategy struggles in choppy markets, since the fluctuations in the ADX cause false signals and never allow the stock to gain sufficient upwards momentum, preventing the strategy from gaining money. The JDM stock increased by 117.76% over the time period, but due to the up-and-down fluctuations of the stock in its gradual rise, strategy 1 sustained losses. This strategy mostly made a profit, and had relatively fine values for its volatility, Sharpe ratio, and max drawdown. Interestingly, it outperformed the other two when investing in stock that experienced declines in value, likely due to its reliance on the ADX preventing it from falling for the same false signals that cost the other two strategies.

The mean reversion strategy (2) performs well in calmer markets with smaller variability, as reversion of the stock to the mean can be counted on more reliably. The same up and down (but gradually increasing and not high variance) nature of the JDM stock which caused strategy 1 to fail caused strategy 2 to see a 38.52% increase, as JDM, despite oscillating up and down, reliably returned to and then exceeded its average stock price. Unsurprisingly, the strategy performs poorly in extremely volatile markets which constantly break through their Bollinger bands, as it faces risks of either catching a falling knife (as a stock falls past the point at which it is expected to recover) or miss out on bigger gains (as a stock rockets past its upper Bollinger band). The first weakness caused the losses in BA, as the stock price fell more than expected, and the second weakness caused the poor performance in investing in the SPY, as bailing out past upper Bollinger bands caused minimal returns. Interestingly, this strategy had both the highest winrate and the largest volatility, suggesting that although its investments were mostly successful, it was prone to ups and downs. This is also seen in its relatively large max drawdown statistics, as well as its low Sharpe ratio, suggesting that the risks associated with this strategy may not be worth any eventual reward.

The volatility breakout strategy (3) performs well when mostly calm stocks experience periods of high growth and volatility, as the combination of the ATR and the Williams %R allow it to predict breakouts as they happen. However, it is prone to false signals as well as periods of market uncertainty, as erratic swings can undermine the careful volatility predictions. While it performed better and more reliably than the other two strategies, it did suffer losses in the XOM stock investment, due to a near constant supply of false signals which added up over the time period. Of the three, it was the most reliable and profitable, and, especially in stocks whose value increased, it produced the strongest Sharpe and Sortino ratios. Despite this, the strategy had a relatively low winrate, almost certainly due to its propensity to fall for false signals. Yet when it read the market correctly, its profits exceeded those of both other strategies, making up for smaller losses.